

RISHI KIRAN KARNATAKAM

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SUMMARY

Highly motivated Computer Science student with a strong passion for user-centered design and crafting intuitive digital experiences. Eager to leverage expertise in interaction design, prototyping, and AI/ML integration to develop impactful products that delight users and solve complex challenges at Google.

EDUCATION

NNRESGI <i>Bachelor of Technology, Computer Science and Engineering</i>	Dec 2021 - Present <i>GPA: 8.4</i>
KMIIT <i>Intermediate, 10+2 equivalent</i>	Sep 2019 - Jun 2021 <i>GPA: 9.4</i>
SAGHS <i>Secondary School Certificate</i>	May 2019 <i>GPA: 9.8</i>

SKILLS

- **Design UX Principles:** User-Centered Design, Interaction Design, Information Architecture, Usability Testing, Prototyping, Wireframing
- **Programming Languages:** Python, C, C#, Dart
- **Machine Learning AI:** TensorFlow, Transfer Learning, Deep Learning
- **Web Mobile Development:** Flask, Flutter
- **Databases:** MySQL, MongoDB
- **Cloud Technologies:** Amazon Web Services (AWS)
- **Tools Version Control:** Git, GitHub, Unity Engine, Tkinter, MongoDB Atlas, Jenkins

PROJECTS

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|---|----------|
| AI-Integrated Plant Disease Detection Mobile App
<i>Flutter, Dart, TensorFlow Lite, MongoDB Atlas</i> | Apr 2025 |
|---|----------|
- Led the design and development of a user-centered mobile application for real-time plant disease detection, enhancing user accessibility to agricultural insights.
 - Integrated a TensorFlow Lite model (Inception V3) for accurate, offline disease prediction, ensuring a seamless user experience even without internet connectivity.
 - Implemented local data storage with automatic syncing to MongoDB Atlas, optimizing data management and reliability for diverse user environments.
 - Designed and crafted an intuitive, modern UI with a comprehensive sidebar navigation, significantly improving user flow for history tracking, profile management, and AI prediction viewing.
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| AI-Powered Chess Engine with Genetic Algorithm Optimization
<i>Python, Tkinter, Chess Library</i> | Aug 2024 |
|---|----------|
- Developed an AI-driven chess engine featuring an interactive graphical user interface (GUI) designed for engaging user play.

- Optimized AI decision-making using minimax, alpha-beta pruning, and Genetic Algorithms, enhancing strategic depth and improving the player's cognitive engagement.
- Achieved a 30% improvement in move generation speed through algorithmic enhancements, contributing to a fluid and responsive user interaction.

Cotton Plant Disease Detection and Classification Using Transfer Learning

Feb 2024

Python, TensorFlow

- Developed a machine learning model for classifying cotton leaf diseases, with a focus on delivering clear, actionable insights for users through visual interpretation.
- Enhanced detection accuracy by 25% through the application of transfer learning with pre-trained CNN models, establishing a robust foundation for user-facing diagnostic tools.
- Achieved 90%+ accuracy on test datasets, optimizing core performance for agricultural disease diagnosis critical for reliable user applications.

Interactive Solar System Model and 3D Game Development

Oct 2023

C#, Unity Engine

- Designed and engineered an interactive 3D solar system model, enhancing gamified learning experiences and fostering deeper user engagement.
- Created a first-person 3D game inspired by Flappy Bird, focusing on intuitive controls, dynamic renderings, and immersive gameplay elements to maximize user enjoyment and retention.

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification, Stanford University (Coursera), Jun 2023
- The Joy of Computing Using Python, IIT Madras (NPTEL), Jul 2023
- Python For Data Science, IIT Madras (NPTEL), Jan 2025
- AICTE Virtual Internship, Nalla Narasimha Reddy Education Society's Group of Institutions, 2024
- AWS Academy Graduate - AWS Academy Cloud Architecting, AWS Academy, Sep 2024

AWARDS

- Internal Hackathon on Generative AI, Achieved 3rd place in a college-hosted Hackathon focused on Generative AI, Aug 2024
- National Hackathon on AR/VR Development, Ranked in the top 10 at a national-level hackathon focused on AR/VR, Oct 2023
- National Hackathon on Generative AI, Achieved 2nd place in a National Level Hackathon focused on Generative AI, Sep 2024